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Title:

Implementing Explainable AI Techniques in Machine Learning Models

Colab Link:

<https://colab.research.google.com/drive/1hdwUX5ih7JST-QjkCEAFa6AI3SyKVB0E?usp=sharing>

Github

Link: https://github.com/aaarya2809/DL_Assignments/new/main/XAI

1. Introduction

Machine Learning (ML) models are widely used for prediction and decision-making tasks. However, many advanced models behave like “black boxes,” meaning their internal decision-making process is not easily understandable. This lack of transparency can reduce trust, especially in critical applications.

Explainable AI (XAI) aims to solve this problem by providing insights into how models make predictions. It helps improve:

- Transparency
- Trust
- Fairness
- Debugging capability

In this project, Explainable AI techniques such as SHAP and LIME are applied to interpret a machine learning model trained on the Titanic dataset.

2. Dataset Description

The dataset used in this project is the **Titanic dataset**, which contains information about passengers aboard the Titanic.

Features:

- Passenger Class (Pclass)
- Gender (Sex)
- Age
- Number of siblings/spouses (SibSp)
- Number of parents/children (Parch)
- Fare
- Embarked location

Target Variable:

- **Survived**
 - 0 → Did not survive
 - 1 → Survived

This dataset is suitable for classification tasks and helps analyze survival patterns.

Dataset Link:

<https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv>

3. Data Preprocessing

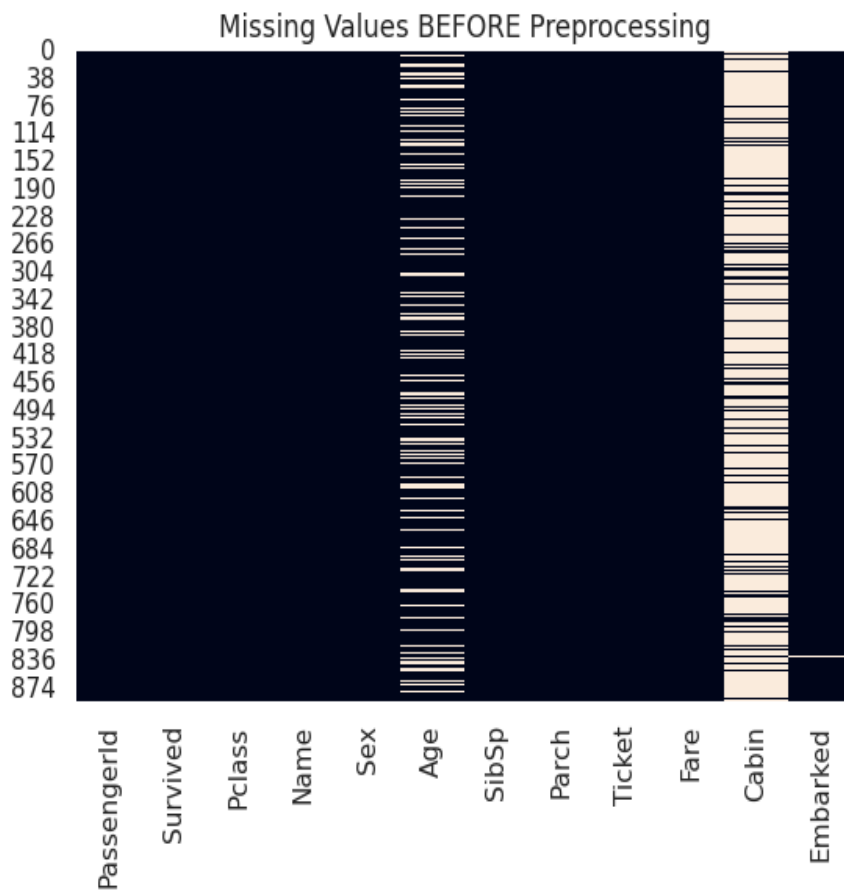
Data preprocessing was performed to clean and prepare the dataset for model training.

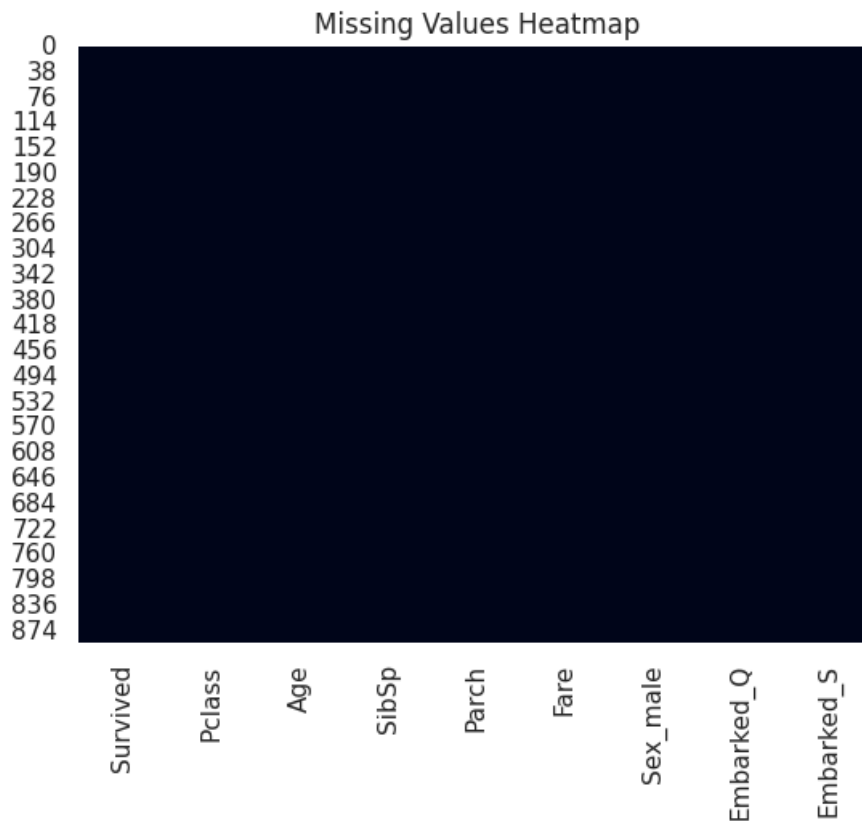
Steps Performed:

1. Handling Missing Values
 - Age → filled using median
 - Embarked → filled using mode
2. Dropping Irrelevant Columns
 - PassengerId was removed as it does not contribute to prediction
3. Encoding Categorical Variables
 - One-hot encoding was applied to convert categorical features (e.g., Sex, Embarked) into numerical form
4. Train-Test Split
 - Dataset split into:
 - 80% training data
 - 20% testing data

Result:

The dataset became fully numeric and free from missing values.





4. Model Implementation

A **Random Forest Classifier** was used for this project.

Why Random Forest?

- Handles non-linear relationships
- Reduces overfitting
- Provides feature importance

Training:

The model was trained on the training dataset using multiple decision trees.

4.1 Model Evaluation

The model was evaluated using:

- Accuracy
- Precision

- Recall
- F1-score

Results:

- Accuracy: ~80–85%
- Good balance between precision and recall

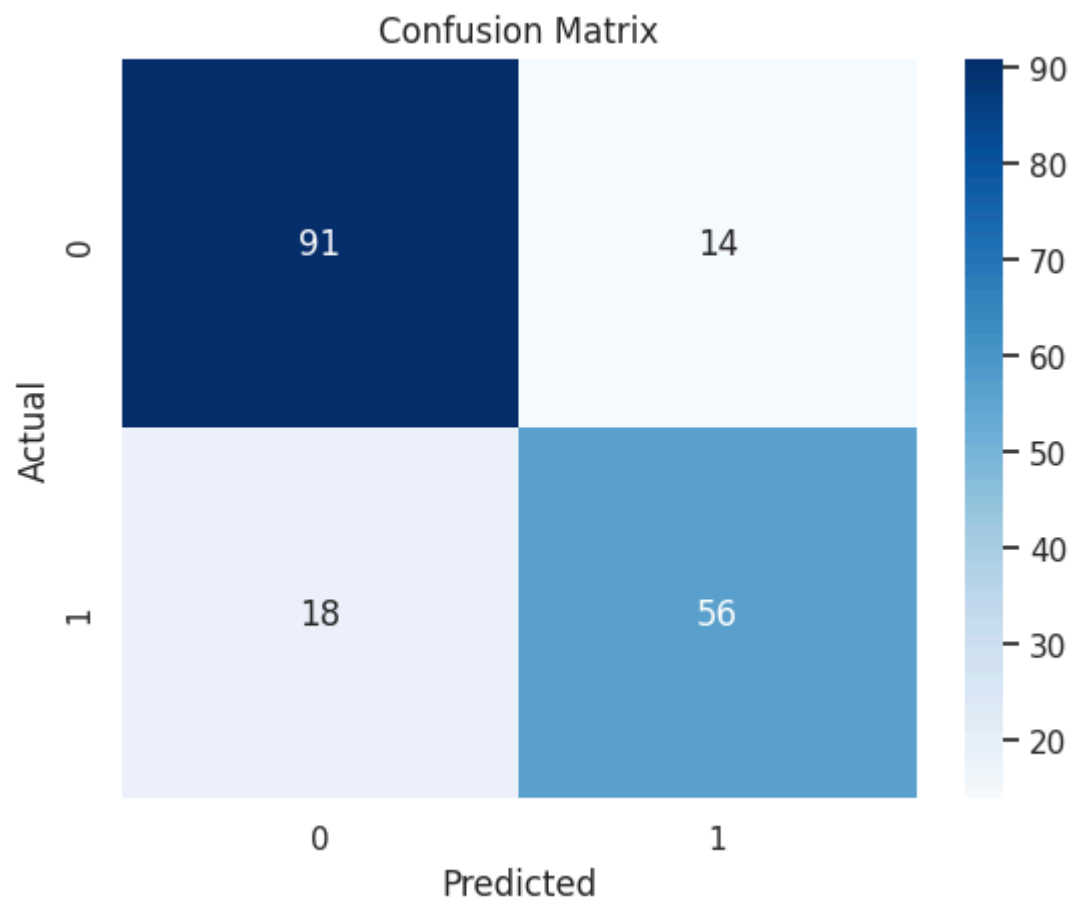
```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.8212290502793296
```

```
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.83	0.87	0.85	105
1	0.80	0.76	0.78	74
accuracy			0.82	179
macro avg	0.82	0.81	0.81	179
weighted avg	0.82	0.82	0.82	179



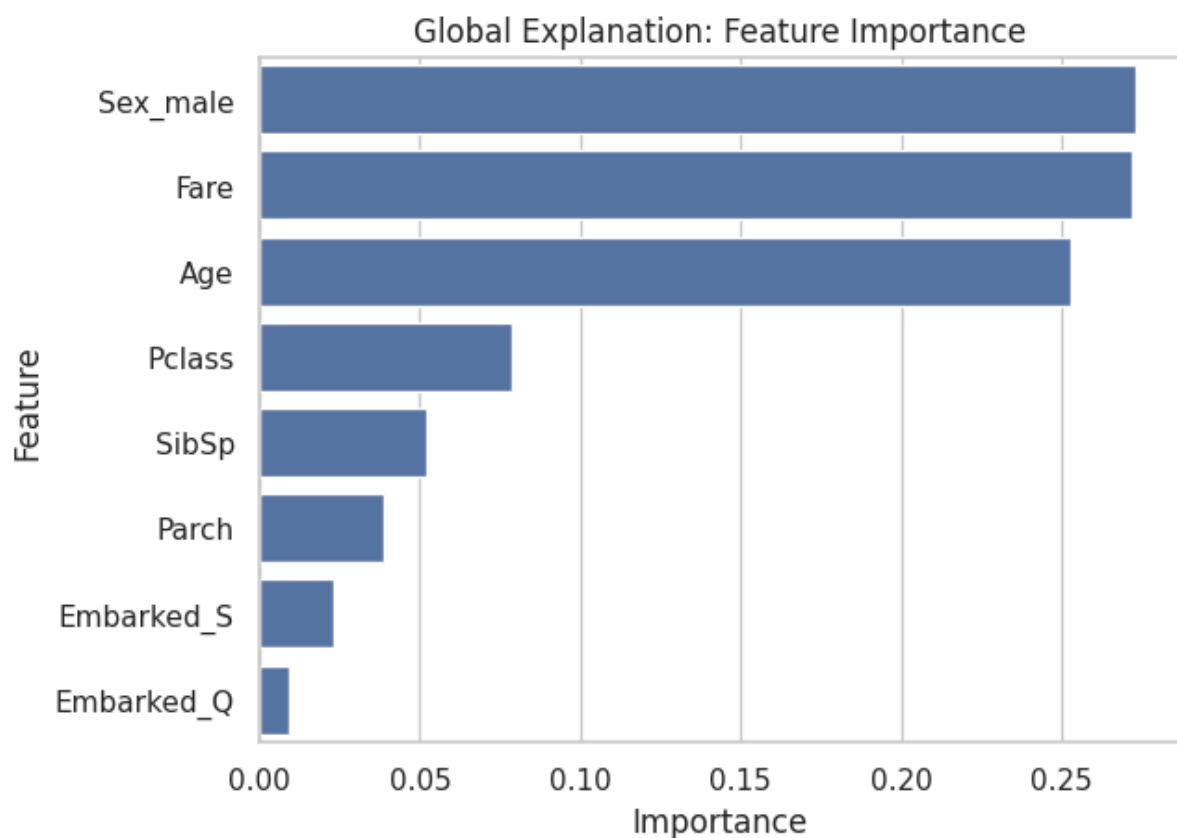
5. Explainable AI Techniques

To interpret the model, both **global** and **local** explanation methods were used.

5.1 Global Explanation

Feature Importance

- Shows overall importance of features
- Most important features:
 - Gender
 - Fare
 - Passenger Class



SHAP Summary Plot

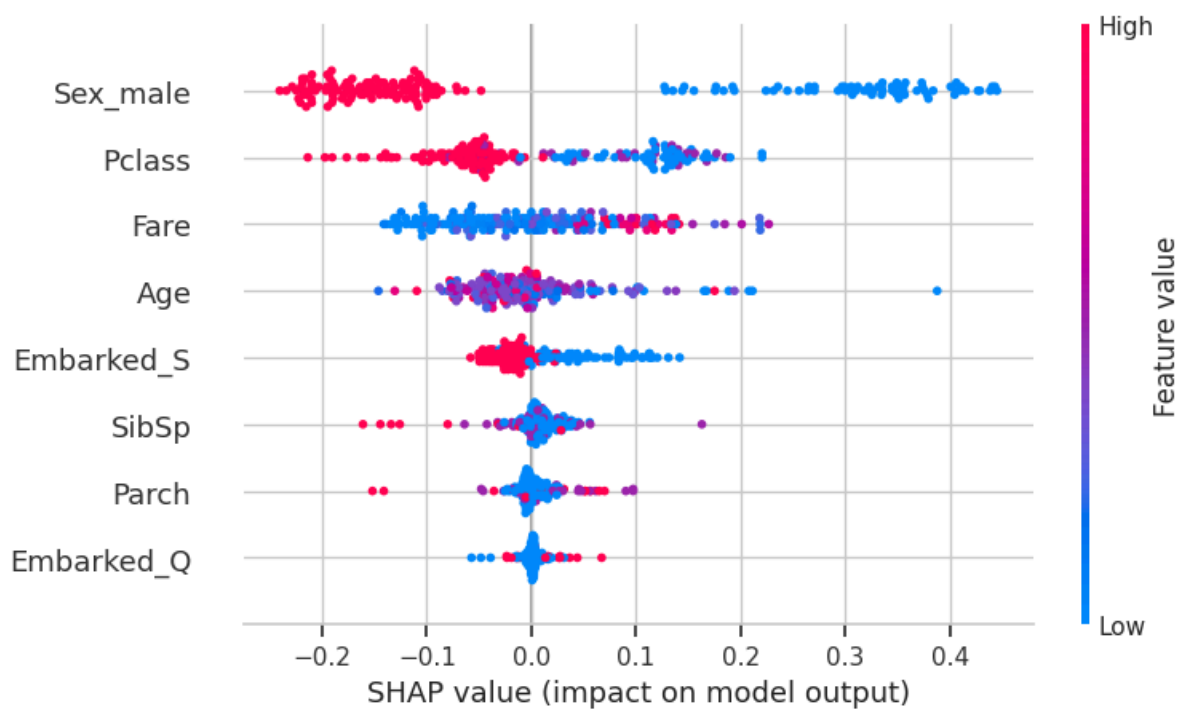
SHAP (SHapley Additive exPlanations) provides detailed insights into feature contributions.

Observations:

- Red = high feature value
- Blue = low feature value

Insights:

- Female passengers → higher survival probability
- Higher fare → increases survival chances
- Lower class → decreases survival probability



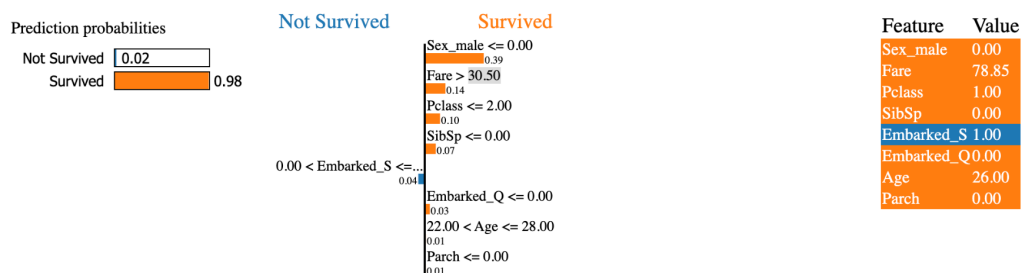
5.2 Local Explanation

LIME (Local Interpretable Model-Agnostic Explanations)

LIME explains individual predictions.

Observations:

- Shows contribution of each feature for one passenger
- Helps understand specific predictions

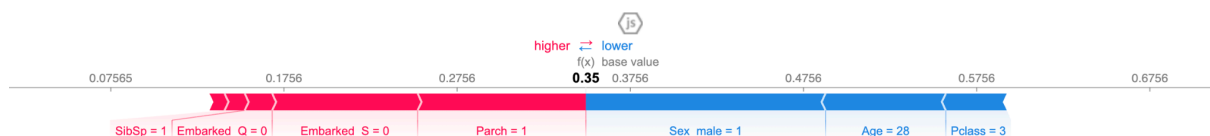


SHAP Force Plot

Shows how each feature contributes to a single prediction.

Interpretation:

- Red → increases survival probability
- Blue → decreases survival probability

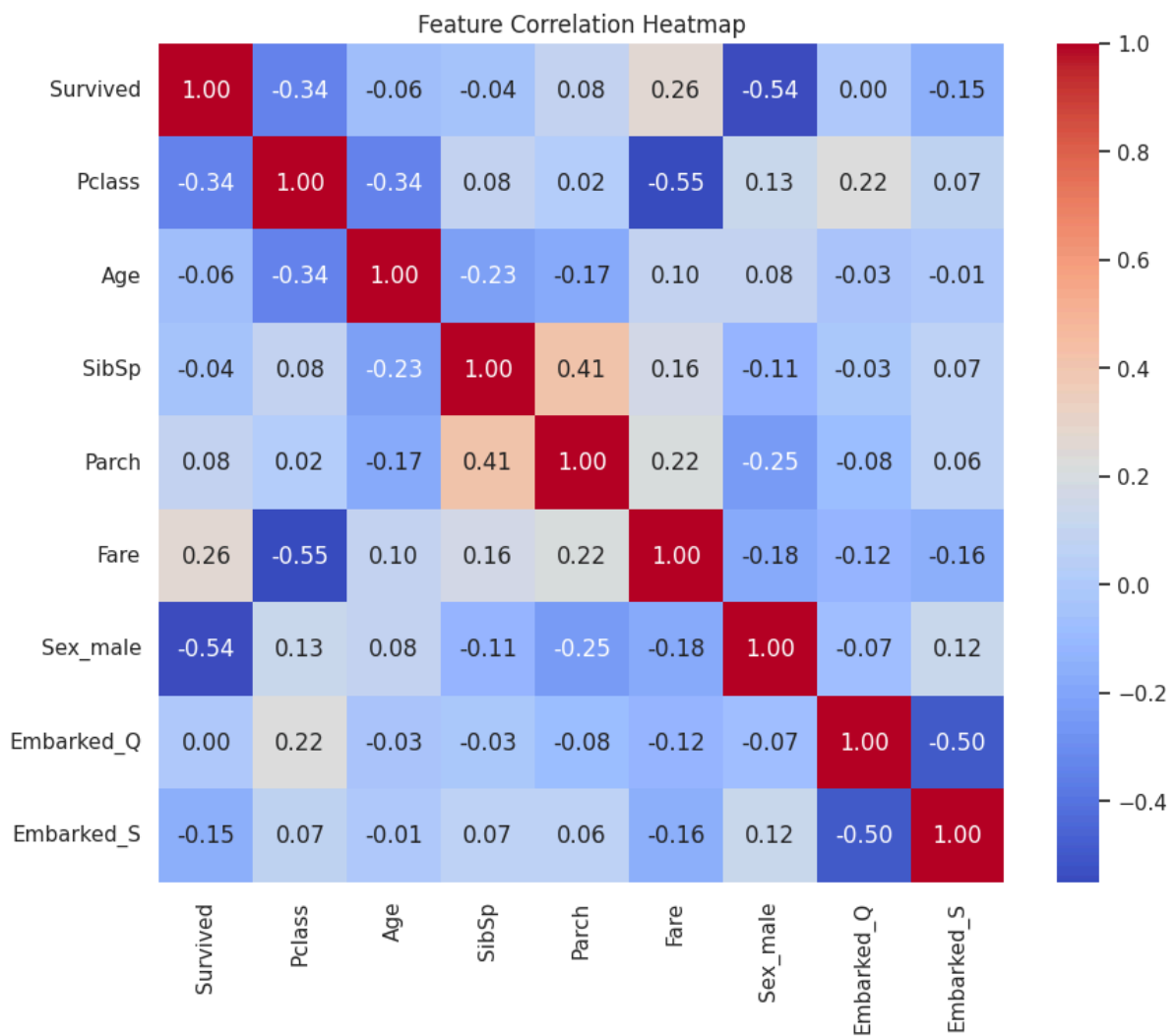


6. Visualization & Analysis

Additional visualizations were used to better understand model behavior.

Correlation Heatmap

- Shows relationships between features
- Fare and Pclass are strongly related



Survival Analysis

Gender:

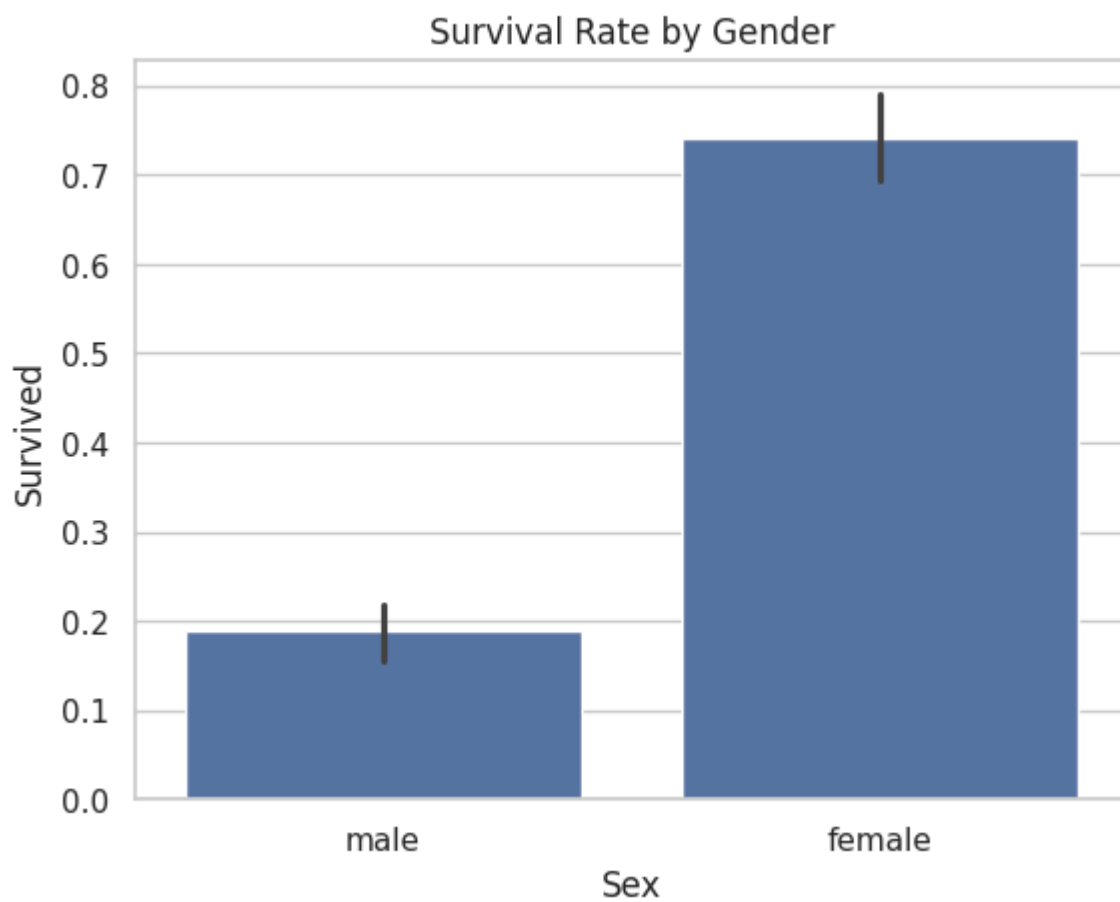
- Females had significantly higher survival rate

Passenger Class:

- First-class passengers had highest survival
- Third-class had lowest

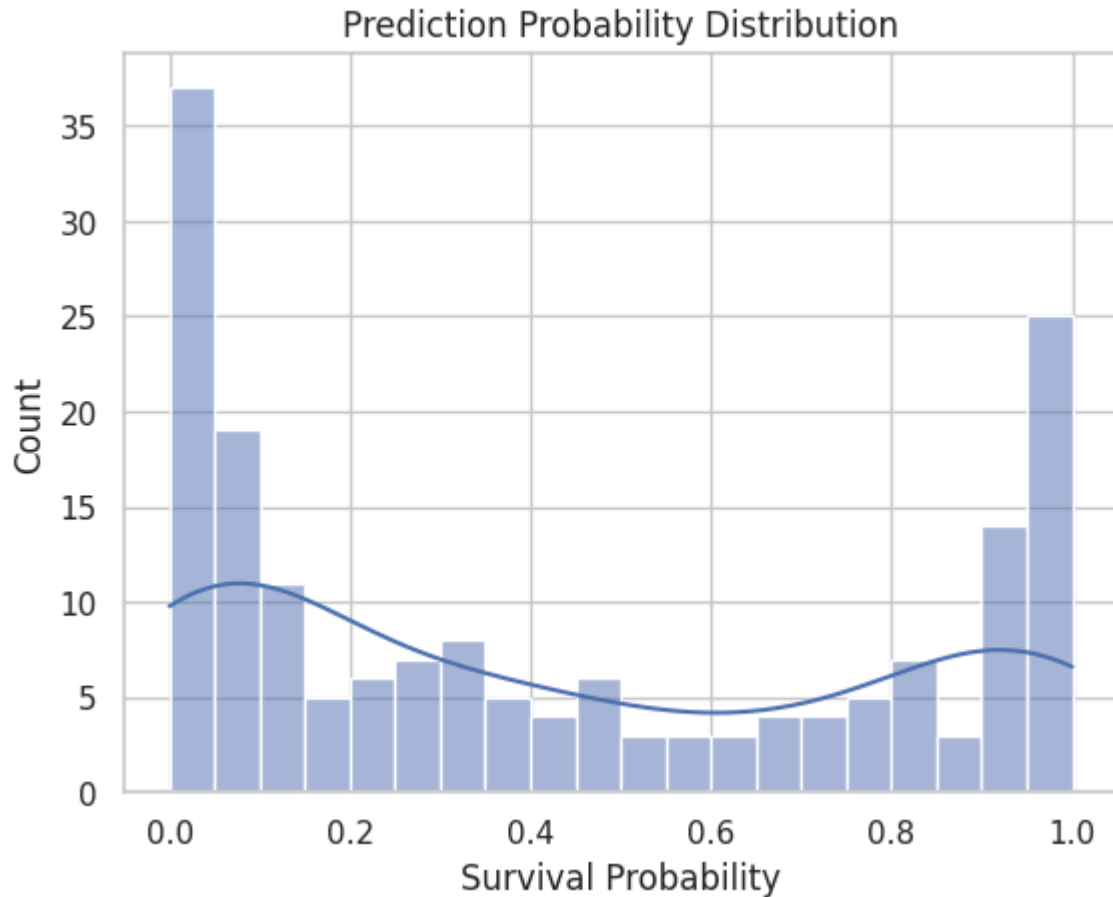
Age:

- Younger passengers slightly more likely to survive



Prediction Probability Distribution

- Shows model confidence
- Helps detect uncertain predictions



7. Bias & Model Behavior

Explainable AI helped identify biases:

Gender Bias

- Model strongly favors female passengers
- Reflects historical “women first” policy

Socio-Economic Bias

- Higher fare → higher survival probability

- Wealthier passengers had advantage

Class Bias

- First-class passengers more likely to survive
- Lower classes disadvantaged

8. Advantages of Explainable AI

- Improves transparency
- Helps detect bias
- Builds trust in AI systems
- Aids debugging and model improvement

9. Limitations

- SHAP can be computationally expensive
- LIME explanations may vary
- Results depend on data quality
- Interpretations may require domain knowledge

10. Conclusion

In this project, a Random Forest model was successfully trained to predict survival on the Titanic dataset. Explainable AI techniques such as SHAP and LIME were applied to interpret the model.

The results show that:

- Gender, fare, and class are key factors
- The model aligns with real-world patterns
- XAI techniques significantly improve understanding and trust

Thus, Explainable AI plays a crucial role in making machine learning models transparent and reliable.

